

Defensive Line Play: Techniques, One-Gap and Two-Gap

Why one great defensive lineman lives in the backfield and another never moves, and how both of them win

Second quarter, first-and-10. The running back takes the handoff, starts toward the right side of the line, and is hit before he gets to it, by a defensive tackle who is already two yards into the backfield. The replay shows the tackle lined up just outside the right guard and gone past him before the guard has finished his first step. “He lives in the backfield,” says the analyst. Two drives later, the same offense runs right up the middle. This time the defender in the center of the line, the biggest man on the field, doesn’t go anywhere at all. The center hits him, a guard comes over to help, and he simply stays where he is, two blockers leaning on him, until the running back arrives and he throws one of them aside and makes the tackle. “He ate up two blockers,” the analyst says. Two defensive linemen, two opposite ways to be great: one wins by *going*, the other by *staying*. This chapter is about why both exist, what each costs, and how to tell which one you are watching.

You’ll be able to:

- Look at a defensive line before the snap and say what each player’s alignment asks of him (nose, 3-technique, 4i, 5-technique, 9, wide-9) and what kind of body the spot calls for.
- Tell one-gap from two-gap play by a lineman’s first step, and say what each style buys the defense and what it gives up.
- Describe how a defensive lineman fights a down block, a reach block and a double team, and why each answer is what it is.
- Explain what it means to set the edge and to keep contain, and find the players doing it.
- Spot a slant or an angle at the snap, and say why the defense moved and where it is now vulnerable.

You’ll need

- **The Language of the Run Game:** gaps (A, B, C, D outward from the center), technique numbers (even numbers head-up on a blocker, odd numbers on his outside shoulder, an “i” on his inside shoulder: a **3-technique** is on a guard’s outside shoulder, a **5** on a tackle’s), *covered* and *uncovered* linemen, *play side* and *backside*, and the blocks this chapter’s defenders have to beat: the down block, reach, cut-off, combo (double team), pull and kick-out.
- **A First Look at Defense:** the 4-3 and 3-4 base defenses (four or three defensive linemen), the box, and edge rushers.

How to read the diagrams. Offense is blue circles, defense orange squares, and plays go up the page. *Left* and *right* always mean the offense’s left and right. LT, LG, C, RG and RT are the five offensive linemen (in the smallest diagrams just T, G and C), Y is a tight end, Q the quarterback and R the running back. Defensive linemen are labelled with their **technique number** (a 3 is a 3-technique); M, W and S are the Mike, Will and Sam linebackers, and B is an outside linebacker standing on the edge. Blue lines ending in a short bar are blocks; orange arrows are where the defender goes; green letters are gaps.

The defensive lineman’s three jobs

Watch a defensive lineman on a running play and it looks like a shoving match he either wins or loses. It is more specific than that. On every snap he has three jobs, in order:

1. **Defeat the block.** A 300-pound man is coming at him, from the front, the side, or two at once. Before he can do anything else, he has to stop that man from moving him where the offense wants him to go.
2. **Control his gap.** Every defense gives each run defender a gap (or two) to be responsible for, so that wherever the ball carrier goes, someone is already there. A lineman who beats his blocker but ends up in the wrong place has won the fight and lost the play.
3. **Find the ball and get to it.** Only after the first two does he get to chase the ball carrier, and by then, on most runs, the ball carrier is already close.

The tension in defensive line play is between the first two jobs. A lineman who is *on* a blocker, hands locked on him, can feel where the block is taking him and can hold his ground, but he is not going anywhere fast. A lineman who is *in a gap*, shooting through the space beside the blocker, can get into the backfield in a heartbeat, but if he guesses wrong about where the ball is going, he has run himself out of the play. Every defense chooses, snap by snap and player by player, which of those two to ask

for. That choice has names, *one-gap* and *two-gap*, and they are the spine of this chapter. First, though, look at *where* the linemen stand, because the alignment is already half of the choice.

The technique map, from the defense's side

In *The Language of the Run Game* you learned the technique numbers as a map the offensive line reads: which blocker has a man on him, which is free. Now turn the map around. To a defensive coordinator, each number is a *job description*, and each job wants a different kind of body.

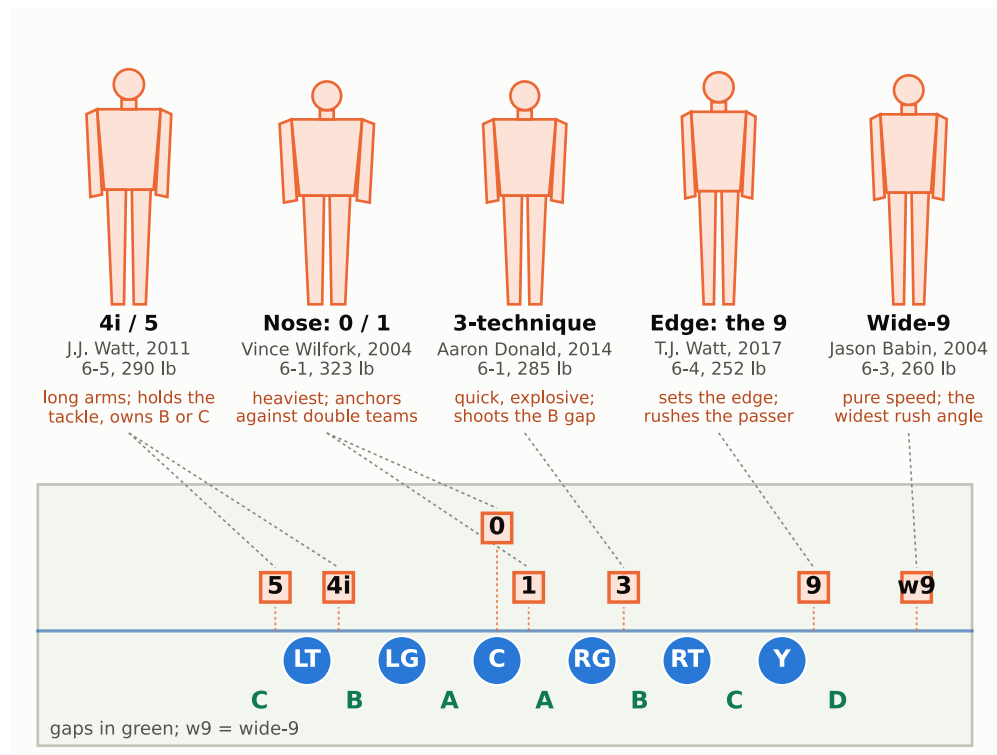


Figure 1: The defensive line's jobs, by alignment, with an NFL archetype for each, drawn to scale from his Scouting Combine height and weight (silhouette width scales with weight per inch of height). The offensive line is drawn with extra-wide spacing so the numbers have room. From left: the 4i and 5 (J.J. Watt, a 5-technique in Houston's 3-4); the nose, at a 0 or a 1 (Vince Wilfork); the 3-technique (Aaron Donald); the 9, outside the tight end, often a stand-up outside linebacker (T.J. Watt); and the wide-9 (Jason Babin, who played it for Jim Washburn in Tennessee and Philadelphia). Same sport, same line of scrimmage: the nose outweighs the edge rusher by about 70 pounds.

Read Figure 1 from the middle out.

The nose: 0 and 1

The man over the center is the **nose**, the nose tackle of The Twenty-Two. He lines up in one of two ways. Head-up on the center he is a **0-technique**, and nothing about his alignment tells the offense which side of the center he will play. On one of the center's shoulders he is a **1-technique**, a *shade nose*, sitting in that side's A gap.

Either way, he is the player offenses most like to double-team. Look at where he stands: the center in front of him and a guard on either side, all within a step. Two of those three can reach him with no effort at all, and the combo block you met in 03-01, center and guard together, is aimed at men in exactly his spot. So the nose's first quality is not speed but *mass and leverage*: the ability to take on two blockers and not be moved. That is why noses are the heaviest players on the field. Vince Wilfork, the model for a generation of them, measured 6-foot-1 and 323 pounds at his Scouting Combine; modern noses such as the Eagles' Jordan Davis (341 pounds) and Dexter Lawrence (342), a Giant until an April 2026 trade sent him to Cincinnati, are just as heavy and much taller.¹

¹Combine heights and weights in this chapter come from the nflverse combine file (`nflreadpy.load_combine()`, which covers 2000 onward), read when the book is built: Vince Wilfork (2004) 6-1, 323; Aaron Donald (2014) 6-1, 285; J.J. Watt (2011) 6-5, 290; T.J. Watt (2017) 6-4, 252; Jason Babin (2004) 6-3, 260; Jordan Davis (2022) 6-6, 341; Dexter Lawrence (2019) 6-4, 342. Players' playing weights changed over their careers; these are their weights as draft prospects. Lawrence's trade from the Giants to the Bengals for the 10th pick of the 2026 draft: NBC Connecticut (April 19, 2026), <https://www.nbcconnecticut.com/news/sports/nfl/giants-trade-dexter-lawrence-bengals-10th-pick-draft/3727122/>.

What does the defense get from a man who absorbs two blockers? Arithmetic. In hat math the offense has a fixed number of blockers. Every blocker spent on the nose is one who isn't climbing to a linebacker, and an unblocked linebacker is a tackle waiting to happen. A great nose's best plays often show up in a linebacker's tackle total, not his own.

The 3-technique: the penetrator

One step outside the guard is the **3-technique**, the alignment that became a position. His spot is the reason. Sitting on the guard's outside shoulder, in the B gap, he is one-on-one with that guard. The center can't easily help (he is two men away), and if the tackle comes inside to double him, the tackle has abandoned whoever was outside *him*. So the 3-technique is the interior lineman most likely to get a single blocker, and the defense puts its quickest, most explosive interior player there to take advantage of it. He is built to get off the ball and shoot the gap before the guard can get his body in the way, on runs and passes alike: Warren Sapp in the 1990s, Aaron Donald in the 2010s, both about 6-foot-1 or 6-foot-2 and listed at roughly 280 to 300 pounds, both much lighter than a nose and much quicker.²

The cost is the mirror image. A 3-technique who is moved off his spot by a double team, or who shoots upfield and misses, leaves the B gap empty. That's the bargain one-gap football makes, and we'll come back to it.

The 4i and the 5: the long men

On the tackle's inside shoulder is the **4i**; on his outside shoulder, the **5-technique**. These are the classic spots of the defensive ends in a three-man front, and the body they want is *long*: tall players with long arms, around 290 to 310 pounds, who can lock out an offensive tackle at arm's length and still see past him to the ball. J.J. Watt, 6-foot-5 and 290 at his combine, played the 5 for the Houston Texans and turned it into a pass-rushing position: he led the league in sacks twice.³

Why the inside shoulder? A 4i sits in the B gap, like a 3, but on the *tackle*. That makes him awkward to block in a way the 3 is not. The tackle can't easily reach him (he is already inside the tackle's body) and the guard has to decide whether to help, so a 4i quietly ties up two blockers' decisions. It is the end's favorite alignment in the five-man *tite* fronts that NFL defenses use to stop inside runs, which you will meet in Odd and Hybrid Fronts. The 5, outside the tackle in the C gap, is the end's spot in four-man fronts as well, where, with no tight end on his side, he is often the last man on the line and has the edge.

The 9, the wide-9 and the edge

Outside the tight end is the 9. Whoever lines up there is usually the defense's edge on that side: in a 4-3 a defensive end, in a 3-4 an outside linebacker standing up rather than in a three-point stance (T.J. Watt in Pittsburgh is the type). He is lighter again, 250 to 270 pounds, because his two jobs both happen in space: rush the passer around the tackle, and on runs, keep the ball from getting outside him. That second job is called setting the edge, and it has its own section below.

Go wider still and you reach the most extreme alignment on the line, the **wide-9**: a defensive end lined up far outside the offensive tackle, outside the tight end or where a tight end would stand, often a good two yards beyond him, usually in a sprinter's stance tilted toward the quarterback. The wide-9 exists for one reason: it gives a fast pass rusher the best possible angle around the tackle. The tight end can't block him without help (he is too far outside), and the tackle has to cover a lot of ground to meet him. The price is just as plain. With the end that far out, the space between him and the next defender inside is enormous, and somebody, usually a linebacker, has to fill it on runs.

²Listed playing sizes: Warren Sapp 6-2, 303 ("Warren Sapp," Wikipedia, https://en.wikipedia.org/wiki/Warren_Sapp; Pro Football Reference lists 300); Aaron Donald 6-1, 280 ("Aaron Donald," Wikipedia, https://en.wikipedia.org/wiki/Aaron_Donald). Donald weighed 285 at his 2014 combine.

³J.J. Watt led the NFL in sacks in 2012 (20.5) and 2015 (17.5) and was Defensive Player of the Year three times (2012, 2014, 2015): Pro Football Reference, <https://www.pro-football-reference.com/players/W/WattJ00.htm>. (Pro Football Reference blocks automated checks; the same years are in "J. J. Watt," Wikipedia, https://en.wikipedia.org/wiki/J._J._Watt.) His role as the starting left end in Wade Phillips's 3-4, "a five-technique between the right tackle and tight end": Houston Texans, "2011 position preview: defensive line" (September 7, 2011), <https://www.houstontexans.com/news/2011-position-preview-defensive-line-5715572>.

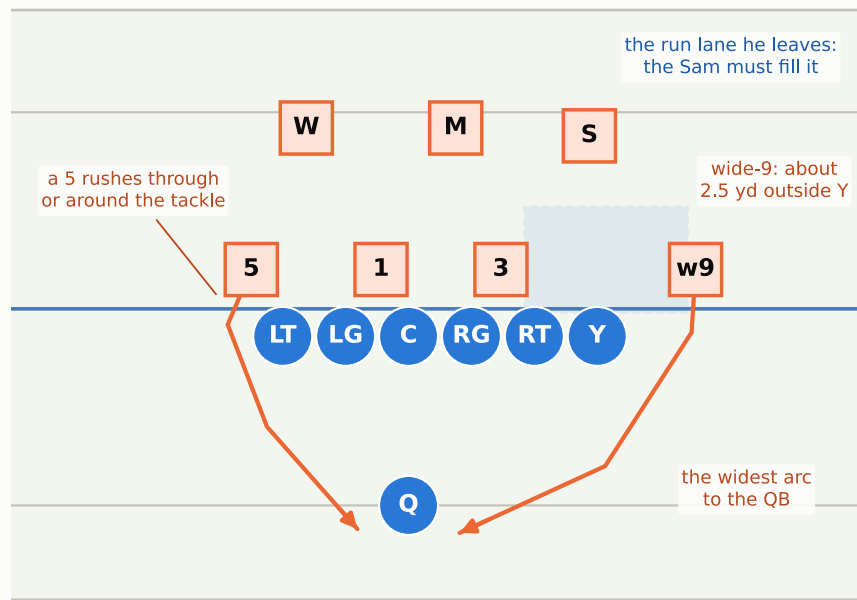


Figure 2: The same four-man line with two kinds of end. Left: a 5-technique on the tackle’s outside shoulder. Right: a wide-9, two and a half yards outside the tight end. His rush path (orange arrow) swings around the tackle on the widest possible arc; the 5’s path has to go through or around a blocker who is right in front of him. The cost is shaded: with the end that wide, the off-tackle lane between the 3-technique and the end is about five yards across, and a linebacker (the Sam) has to fill it on a run.

The wide-9’s modern history runs through one coach. Jim Washburn, the Tennessee Titans’ defensive line coach from 1999 to 2010 and the Philadelphia Eagles’ in 2011 and 2012, built his lines around ends who lined up wide and attacked upfield on every snap. Washburn had a reputation for blunt talk, and he described his philosophy in a sentence: “I like fast guys that play hard and attack.” Gap assignments, he said, were not the point: the job was to “whip the blocker and tackle the ballcarrier if he’s in that gap or that gap. Just play football.”⁴ In his first Philadelphia season the Eagles tied for the league lead with 50 sacks, and Jason Babin, whom Washburn had also coached in Tennessee, had 18 of them.⁵ Jim Schwartz, the Titans’ defensive coordinator for most of Washburn’s years there (2001 to 2008), took the same wide, attacking front to his later jobs, including the Eagles team that won Super Bowl LII and the Cleveland defenses of 2023 to 2025 in which Myles Garrett was twice Defensive Player of the Year and set the single-season sack record of 23 in 2025. (Those Cleveland fronts were billed as wide-9 when Schwartz arrived, but no modern defense lines up that way on every snap: how wide the ends set changed with down, distance and personnel.)⁶ How the wide-9 fits into a whole four-man front, and what it does to the linebackers behind it, is in Even Fronts.

! Why it exists

Every alignment is a trade between the run and the pass, and between size and speed. Inside, where blockers are close together and double teams are easy, the defense wants mass that can’t be moved: the nose. One step outside, where a defender can get a single blocker, it wants quickness: the 3-technique. On the tackle, it wants length: the 4i and 5. On the edge, it wants speed, and the wider it aligns its edge rusher the more pass rush it gets and the more run defense it gives up. When you read a front, read it as a list of bets. Which spots did the defense fill with which kind of player, and what is it hoping you’ll do?

⁴Washburn’s coaching career (Titans defensive line coach 1999–2010, Eagles 2011–2012): https://en.wikipedia.org/wiki/Jim_Washburn. Quotations: Philadelphia Eagles, “Washburn waxes on scheme, personnel” (February 9, 2011), <https://philadelphiaeagles.com/news/washburn-waxes-on-scheme-personnel-3636698>.

⁵Philadelphia Eagles, “Position Preview: Defensive End” (January 17, 2012): “the Eagles tied for the league lead with 50 sacks,” and “Babin finished the season third in the league with 18.0 sacks”: <https://www.philadelphiaeagles.com/news/position-preview-defensive-end-6825622>.

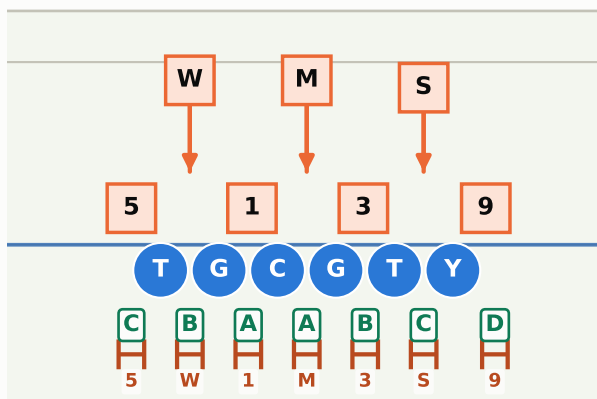
⁶Jim Schwartz: Titans defensive coordinator 2001–2008, Eagles defensive coordinator 2016–2020 (Super Bowl LII, after the 2017 season), Browns defensive coordinator 2023–2025; he resigned in February 2026: https://en.wikipedia.org/wiki/Jim_Schwartz. Myles Garrett’s 23.0 sacks in 2025 (the single-season record) and his unanimous 2025 Defensive Player of the Year award: NFL.com, <https://www.nfl.com/news/browns-de-myles-garrett-reaches-23-sacks-sets-new-nfl-single-season-record>, and this book’s checked facts (planning/FACTS-current.md §2). Garrett was also 2023 Defensive Player of the Year. The wide-9 in Cleveland: Anthony Poisal, “Jim Schwartz aims to help Myles Garrett play ‘free and more productive,’” ClevelandBrowns.com (January 18, 2023), <https://www.clevelandbrowns.com/news/news-notes-jim-schwartz-aims-to-help-myles-garrett-play-free-and-more-productive>.

One-gap: attack a gap

Now the big choice. In a **one-gap** defense, each defensive lineman is responsible for exactly one gap. He lines up in it or shaded toward it (that is what an odd number or an “i” usually signals), and at the snap he attacks it. His goal is **penetration**: getting across the line of scrimmage and into the offense’s backfield before the play can develop, through his gap. If a blocker gets in his way, he defeats the blocker *while staying in his gap*; he does not need to read where the ball is going first, because the other gaps belong to somebody else.

That last clause is what makes one-gap football work: **the gaps add up**. Seven gaps across a line with one tight end, seven defenders in the box, one defender for each. The linemen take the gaps they are aligned in; the linebackers behind them take the rest. Everybody has one job and can play it fast.

One-gap 4-3: seven gaps, seven men



Two-gap 3-4: three men, six gaps

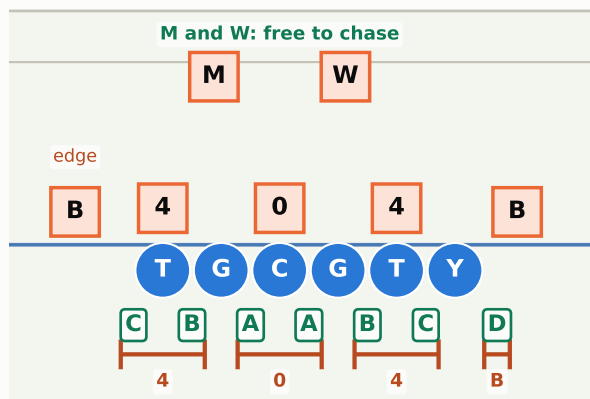


Figure 3: Who owns which gap, against the same offense (one tight end, so seven gaps: two A, two B, two C and a D on the tight-end side). The row under the gap letters says who owns each one. Left, a one-gap 4-3: each of the seven defenders in the box has one gap. The linemen take the gaps they are shaded into (5, 1, 3, 9) and the three linebackers (arrows) fill the three left over. Right, a two-gap 3-4: the nose (0) and the two ends (4, head-up on the tackles) each own the two gaps on either side of their blocker (the wide bars), and the outside linebacker on the right takes the D gap. Three players cover six gaps, so the left outside linebacker can play the edge and both inside linebackers are free to run to the ball. This is the basic accounting; real defenses add wrinkles to both pictures.

Here is a one-gap 3-technique doing his job. The offense runs outside zone to the right, the run from Zone Running in which every lineman steps to the play side and tries to get his head outside the defender in front of him. For the right guard, that means *reaching* the 3-technique: getting from the 3’s inside shoulder to his outside shoulder, against a man who is already outside him. The 3-technique’s answer is simply to be faster.

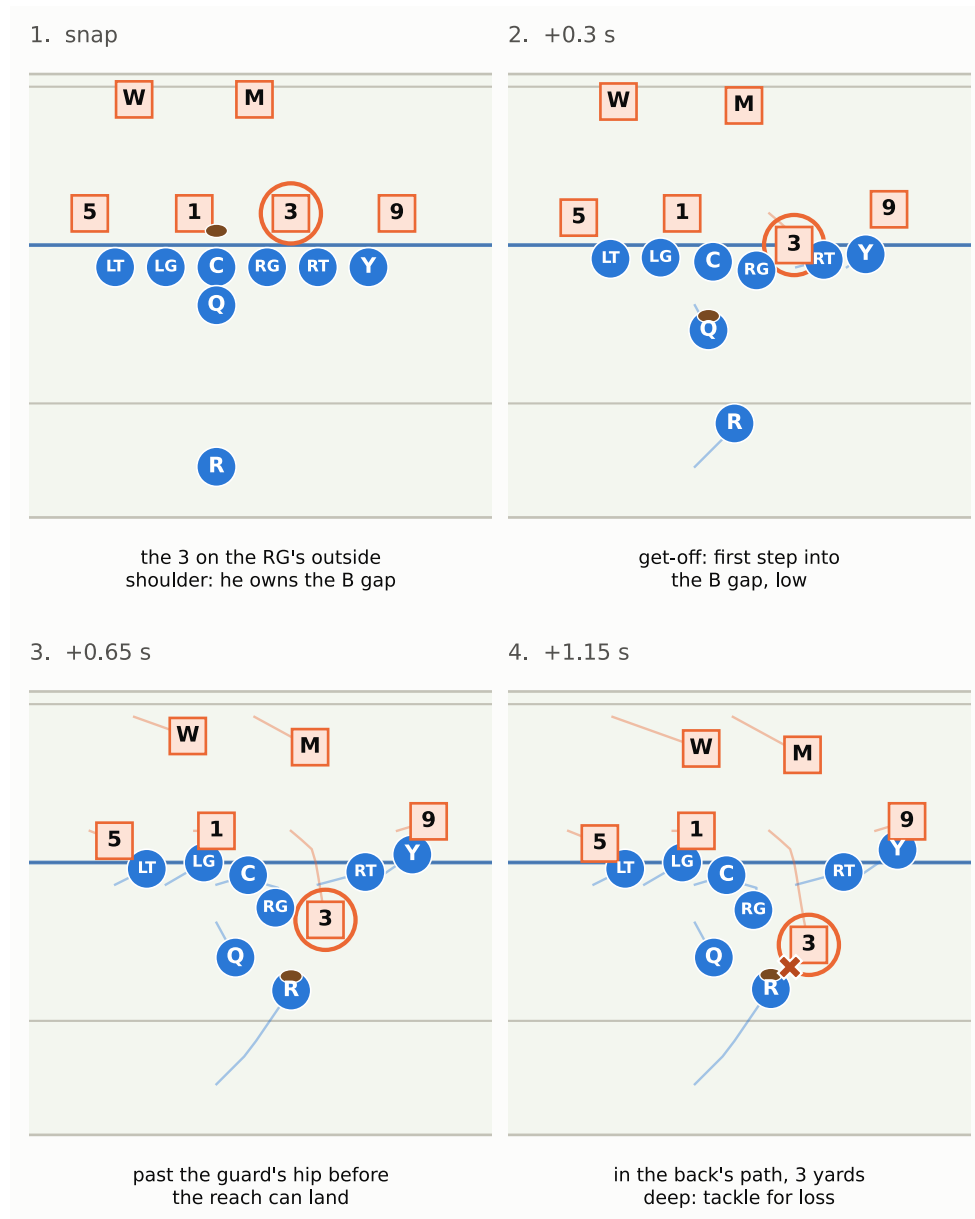


Figure 4: A one-gap 3-technique against outside zone to the right (simulated, in Big Data Bowl tracking coordinates; faint lines are each player's path so far). 1: the 3 (ringed) is on the right guard's outside shoulder; his gap is the B gap. 2: the get-off. He moves on the ball, not the snap count, and his first step is already into the B gap while the guard is still stepping sideways to reach him. 3: he is past the guard's hip before the guard can get his head across: penetration. 4: he is in the running back's path three yards behind the line, and the back, who wanted to run outside the tight end, is caught before he can turn upfield (X).

Why it exists. One-gap play turns the defense's linemen loose. A lineman who doesn't have to read before he moves gets a half-step head start on every snap, and on the line half a step is everything: the blocker who is reaching or pulling or setting for a pass is moving *sideways* or *backward* while the defender is moving forward. The results are the plays fans remember, tackles for loss, sacks, a running back hit as he takes the handoff, and they wreck an offense's plans in a way a two-yard gain never does. One-gap play also fits the players the modern game produces. A lineman who wins by quickness, the Donald type, is wasted standing still with his hands on a blocker; turn him loose in a gap and he is the most disruptive player on the field.

What it costs. Three things.

- **A wrong guess is a big hole.** A one-gapper who penetrates *past* the play, or gets moved out of his gap by a double team, leaves his gap empty, and nobody else was assigned to it. A run that gets through a vacated gap is usually not a short gain, because the next defender is a linebacker who was busy with his own gap.

- **Offenses build plays to punish it.** An aggressive penetrator is exactly what the *trap* in Gap and Power Runs is designed for: let him into the backfield unblocked, then hit him from the side with a pulling lineman, and run the ball through the space he left. Draws and screens work on the same idea: invite the rush, then go where it was.
- **It needs everyone else to be right.** One-gap defense is only as good as its accounting. If one linebacker fills the wrong gap, there are two defenders in one gap and none in another. That is why one-gap teams spend so much practice time on *run fits*, the subject of Run Fits and Box Math.

Where it came from. For much of football's history, defensive linemen were taught to read and react. The great attacking interior linemen changed that by being too quick to keep at home. The best known story is Pittsburgh's in 1974: late that season the Steelers began lining Joe Greene up at the center, his stance tilted at about 45 degrees toward the A gap, so he could knife through before anyone could block him. Accounts differ on whose idea it was: the Hall of Fame credits Greene, other histories credit his line coach, George Perles. The Steelers called it the *stunt 4-3*. It began as a practice-field experiment, got its first heavy use against New England in late November, and carried them through the 1974 playoffs (O.J. Simpson ran for 49 yards against it) and their first Super Bowl win.⁷ Two decades later the Tampa Bay Buccaneers of Tony Dungy and Monte Kiffin built a whole defense around a one-gap 3-technique, Warren Sapp, who was the NFL's Defensive Player of the Year in 1999.⁸

Two-gap: control the man, then pick a side

The other answer is older, heavier and quieter. In a **two-gap** defense, a lineman lines up head-up on a blocker (a 0, a 2, a 4: the even numbers) and is responsible for the gaps on *both* sides of him. At the snap he does not attack a gap. He attacks the *man*: strikes him with both hands inside the blocker's chest, straightens his arms to keep the blocker off his body (coaches call it *locking out*), and stands him up at the line of scrimmage. Then, with the blocker held at arm's length, he looks into the backfield, sees which side the ball is going, and throws the blocker the other way to make the tackle.

⁷Bob Labriola, "Steelers begin run to Super Bowl IX," Steelers.com (November 26, 2014): the Stunt 4-3 "put Greene directly over the center but in a stance at a 45-degree angle"; it began as "an experiment confined to Thursdays on the practice field" and was used "23 times in 60 snaps against the Patriots"; Simpson ran for 49 yards on 15 carries in the 32-14 divisional playoff win: <https://www.steelers.com/news/steelers-begin-run-to-super-bowl-ix-14412613>. The Pro Football Hall of Fame says Greene "developed the new tactic of lining up at a sharp angle between the guard and center": <https://www.profootballhof.com/players/joe-greene/>. "George Perles," Wikipedia, calls the Stunt 4-3 "designed by Perles": https://en.wikipedia.org/wiki/George_Perles.

⁸Warren Sapp: Buccaneers 1995-2003, Defensive Player of the Year 1999, 96.5 career sacks, Hall of Fame class of 2013: <https://www.profootballhof.com/players/warren-sapp/>; https://en.wikipedia.org/wiki/Warren_Sapp. Dungy was Tampa Bay's head coach 1996-2001 and Kiffin its defensive coordinator 1996-2008: "Tampa 2," Wikipedia, https://en.wikipedia.org/wiki/Tampa_2.

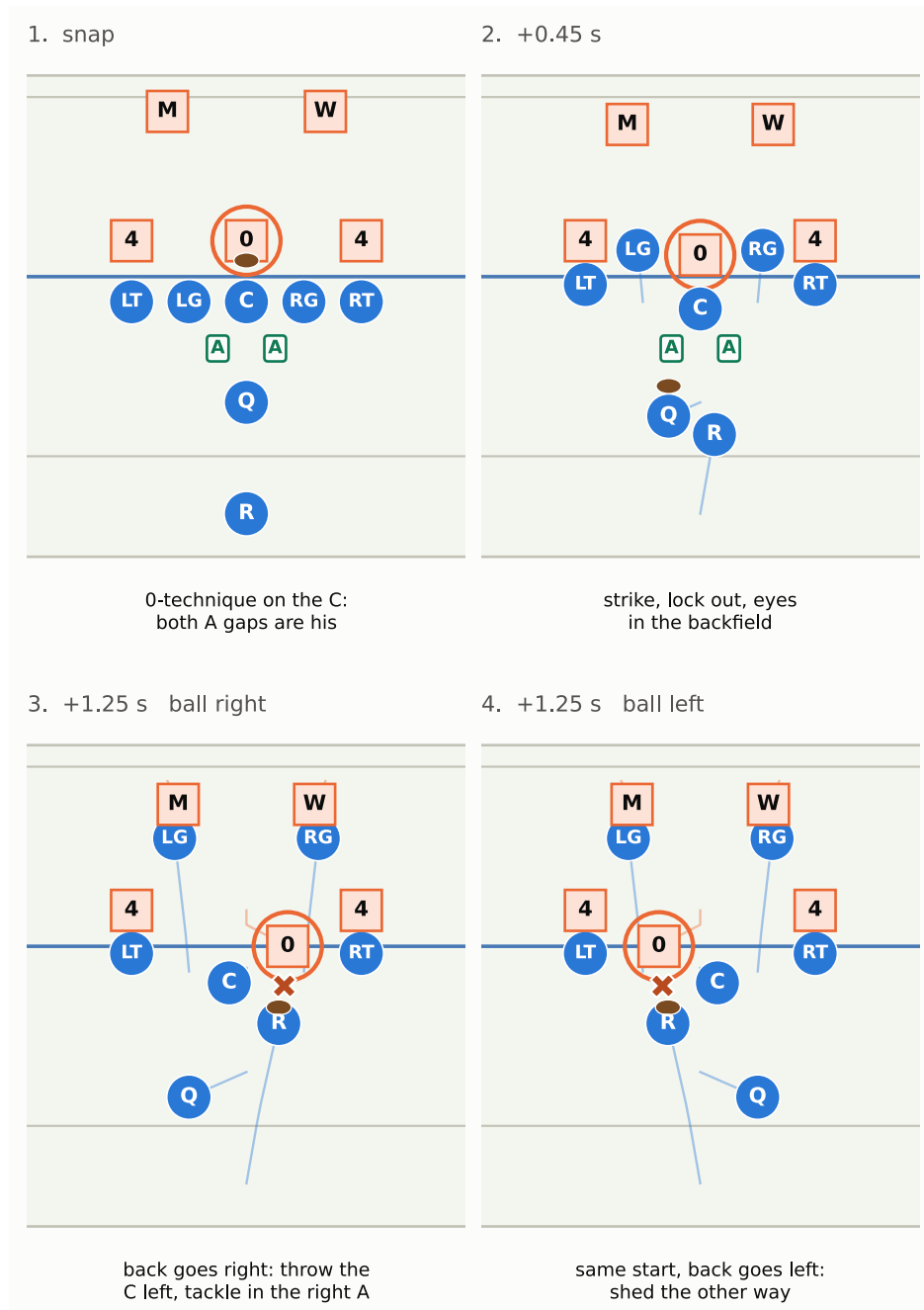


Figure 5: A two-gap nose tackle in a 3-4, the same start with two endings (simulated, Big Data Bowl coordinates). 1: the nose (ringed) is a 0-technique, head-up on the center, responsible for both A gaps (green). The guards are uncovered and will climb to the inside linebackers. 2: the strike. Both hands inside the center's chest, arms locked out, the center held a yard off his body and pushed back a step; the nose's eyes are on the running back, and both A gaps are still his. 3: the back commits to the right A gap, so the nose throws the center to his left and steps into that gap for the tackle (X). 4: the same play, but the back cuts to the left A gap, so the nose sheds the other way. The offense could not know in advance which side he would take: that is the point.

Coaches have a rule for how a two-gapper knows which way to go: **read the blocker's head**. A blocker who wants to wall you off from one side has to get his helmet to that side of you. If the center's head slides to your left, he is trying to keep you from going left, which means the ball is going left. So the two-gapper "plays across the face": he keeps his arms locked, works to the side the blocker's head went, and throws him the other way. The block tells him where the ball is going before he ever sees the ball.

Why it exists. Look back at the right half of Figure 3. If one lineman can be responsible for two gaps, three linemen can hold six of them, and the defense has players left over. In the classic two-gap 3-4, the two inside linebackers have no gap of their own to rush to at the snap; they read the play and run to the ball, unblocked, because the big men in front of them are occupying all five

offensive linemen. That is the two-gapper's real product: not tackles, but *clean linebackers*. And because he doesn't commit before the snap, the offense can't build its blocking around where he will go.

Two-gapping matters even more in the modern NFL than its reputation suggests. Defenses now keep two deep safeties on many snaps to stop the pass, which leaves only six men in the box against an offense that may have six or seven blockers (A First Look at Defense has the numbers). With six defenders and seven gaps, the accounting of one-gap football does not add up. The usual fix is to ask a big nose to play both A gaps, so six men can cover seven gaps. That is one of the ideas behind the light-box fronts of Vic Fangio and his followers, taken up in Odd and Hybrid Fronts and the case study Fangio's Two-High Revolution and Macdonald's Answer.

What it costs.

- **Rare bodies.** Holding a 310-pound center at arm's length while a guard hits you from the side takes size and strength that very few people have. Two-gap noses are typically well over 320 pounds, and the pool of such players who can also move is small.⁹
- **No head start.** Reading before moving means the two-gapper is always reacting. He rarely makes a tackle behind the line and rarely gets a sack; on passing downs a read-first lineman gives the quarterback time. That is why classic two-gap noses often leave the field on third down for a nickel package.
- **His value hides.** The stat sheet credits the man who makes the tackle, not the man who made the tackle possible. A great two-gapper can have a quiet box score in a game he dominated.

Common misconception

"A 3-4 means two-gap, and a 4-3 means one-gap." It is a decent first guess, nothing more. The classic 3-4s of the 1970s through the 2000s, Bill Parcells's and Bill Belichick's among them, did two-gap with big ends and a big nose. But Wade Phillips, son of the Bum Phillips who devised the technique numbers, spent decades running an *attacking*, one-gap 3-4: when he took over the Houston Texans' defense in 2011, the team's own preview called it "his one-gap scheme," and the line coach explained that it was "not like a two-gap type defense where you need that 350-pound nose man to be able to just clog up and take on two blocks."¹⁰ And plenty of 4-3 teams ask a nose to two-gap. The number of linemen tells you how many there are; their first step tells you how they play.

Madden vs. real life

In Madden, a defensive lineman is mostly someone you steer at the quarterback, and the run defense takes care of itself. On a real early down, a lineman's job is mostly the gap: hold it, or attack it, exactly as the call says. A nose tackle who abandons both A gaps to chase a sack can give up a 20-yard run even if he beats his man, which is why defensive line coaches grade every snap on assignment before they grade it on result.

Fighting the blocks

Whichever style he plays, a defensive lineman has to beat whatever block the offense throws at him, and the offense has a whole vocabulary of them. You learned them from the blocker's side in 03-01. Here are the defender's answers to the three that matter most, the down block, the reach and the double team, and the one principle underneath all of them: **fight pressure with pressure**. Whatever direction a block is pushing you, push back into it. A blocker pushes you *where the offense wants you*, toward the hole's edge or out of your gap, so leaning into the pressure is how you stay where the defense wants you.

⁹Pro Football Focus, "Defensive Prototypes: 0-Technique" (the nose's job "to control the center, often draw a double team from a guard," with two-gap noses "usually north of 330 pounds"): <https://www.pff.com/news/defensive-prototypes-0-technique>.

¹⁰Houston Texans, "2011 position preview: defensive line" (September 7, 2011): Phillips "views them as a perfect fit for his one-gap scheme," and defensive line coach Bill Kollar: "We're more still of a one-gap type defense": <https://www.houstontexans.com/news/2011-position-preview-defensive-line-5715572>. Bum Phillips and the technique numbers: The Language of the Run Game.

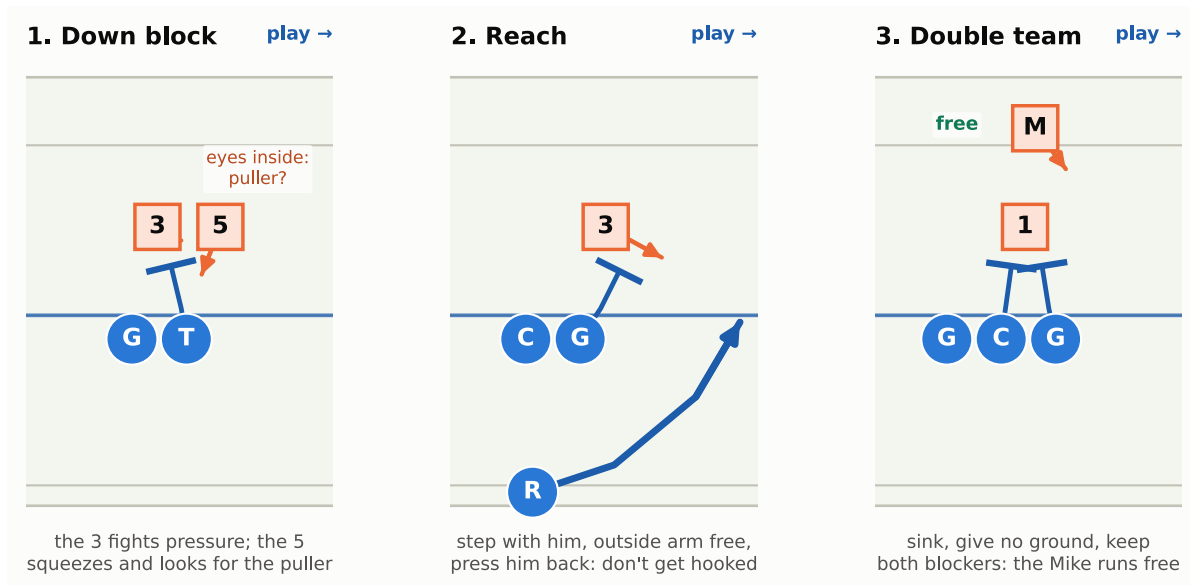


Figure 6: The defender's answers, play going right in every panel (defenders drawn a little deeper than real so each path shows; blue bars are blocks, orange arrows the defender's response). 1, Down block: the right tackle blocks down on the 3-technique. The 3 fights the pressure, pushing back outward so he isn't washed inside, and the 5, left without a blocker, squeezes down the line, hip to hip with the tackle, to shrink the hole while he looks inside for a pulling lineman or a fullback coming to kick him out. 2, Reach: the guard tries to get his head outside the 3. The 3 steps with him, keeps his outside arm free and presses him back toward the backfield, so the runner has to go even wider; he does not run around the block. 3, Double team: the center and guard combine on the 1-technique. He sinks his hips and fights the pressure of the drive man, gives no ground, and keeps both blockers on him, so the Mike behind him is free to fill.

Against a down block

A down block hits a defender from the side: the blocker outside him comes inside, at an angle, to drive him toward the center and seal the inside of the hole. Two defenders are involved, and they have opposite jobs.

- **The man being down-blocked** (the 3-technique in panel 1) feels pressure from his outside. If he gives way, he is washed down the line and the hole doubles in size. So he fights pressure: drops his outside shoulder into the blocker and pushes back outward, holding his gap. If he can do more, some defenses teach him to fight *across* the blocker's head and get into the hole himself.
- **The man left alone** (the 5 in panel 1) has just seen the tackle in front of him leave. A blocker who vanishes is a warning, not a gift. On a gap-scheme run like power, the tackle blocks down *because* someone else is coming for the end: a pulling guard or a fullback, to kick him out. So he **squeezes**: steps down the line, hip to hip with the down-blocker, shrinking the hole, with his eyes inside for the puller. When the puller arrives, the defense's call decides what he does next. He can keep his outside arm free and force the ball back inside (*box* or *squeeze*), or attack the puller's inside shoulder and pile up the hole, pushing the ball outside to a teammate waiting there (*spill*, the "wrong-arm" you met in 03-03). Which one, and who is waiting, is the heart of Run Fits and Box Math.

There is a cousin of this rule that every defensive line coach teaches: **if nobody blocks you, something is coming**. A lineman who is let through untouched should close inside and get small at once, because the likeliest explanation is a trap block from the side, the gap play built to catch penetrators.

Against a reach

A reach block is the blocker trying to get his head and body *outside* a defender who is already a step to the play side of him, and seal him inside. The defender's whole job is not to let that happen, and he has the head start. He steps with the blocker, keeps his *outside* arm free (so there is always an arm between the blocker and the sideline), and presses the blocker back into the backfield. Panel 2 shows the effect: the runner, who wanted to turn upfield just outside the guard, has to keep running sideways. Every yard he goes sideways is a yard of pursuit for the rest of the defense.

The mistake to avoid is the one that looks most athletic: running *around* the block. A defender who loops around the blocker's back to get outside him has left his gap. On outside zone, that is exactly the hole the runner is looking for when he cuts back. The rule coaches use is to beat the block *through* the blocker, not around him.

Against a double team

The double team, or combo block, is the hardest thing a defensive lineman faces and the most important to survive. Two blockers want two things from him: to move him off the ball, and then, once the linebacker behind him commits, for one of them to come off and block that linebacker. The defender's goal is to deny both.

- **Don't get moved.** He widens his base, sinks his hips, and fights the pressure of the blocker who is driving him (usually the one head-on). Giving up a yard backward is how a double team creates the hole.
- **Keep both blockers.** The longer he keeps both men busy, the longer the linebacker behind him goes unblocked. The double team's second blocker is supposed to climb to the second level; a defender who stays heavy on him keeps him from leaving.
- **If he is losing, make a pile.** A lineman being driven back is taught to drop his hips, even to go to a knee in the gap, so that the bodies pile up where the hole was supposed to be. It looks like losing. It is a win if the running back has nowhere to go.

A few great linemen *split* double teams, slipping between the two blockers into the backfield, and it is a spectacular play when it works. It is also a gamble. If he doesn't get through, he has left his gap and the double team has moved nobody, because there is nobody left to move.

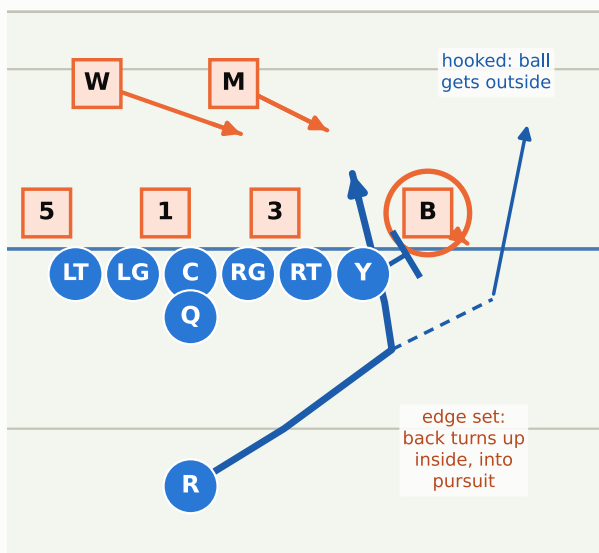
Against blocks going away

When the play goes away from a defender, his blocker uses a cut-off block (or the whole backside line *scoops*), stepping toward the play to get between him and the ball. Here he does chase. He flattens down the line, staying at about the depth of the ball, so that if the runner cuts back he is waiting. But the defender farthest from the play has a different rule, and it comes next.

Setting the edge and keeping contain

The last man on the line on each side has a special job. Inside him, his teammates can count on someone being in the gap next to them. Outside him there is nobody, just the sideline. If a run gets outside him, it gets outside everybody.

Setting the edge



Keeping contain

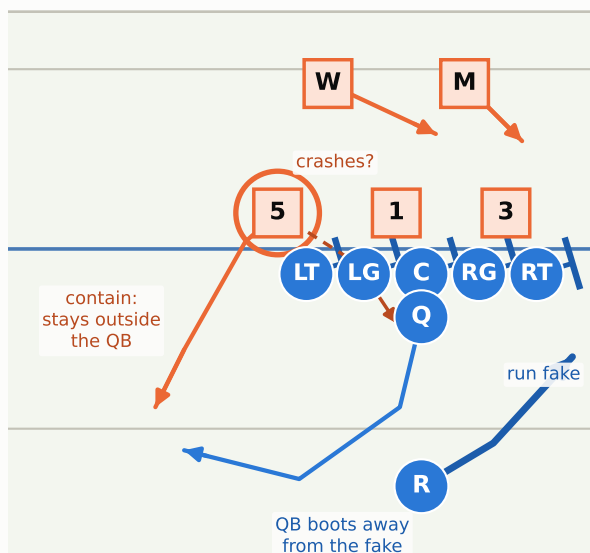


Figure 7: The edge defender's two jobs. Left, setting the edge against a run to the right (outside zone or a toss): the tight end tries to reach the 9-technique outside linebacker (B, ringed). The B steps with him, keeps his outside arm free, and drives him back into the backfield, so the runner cannot get outside and has to turn up inside (the solid path), back toward the Mike and Will chasing from inside. The dashed path is what happens if he is hooked: the ball gets outside everybody. Right, contain against a play-action bootleg: the run fake goes right and the quarterback rolls out left. The backside end (5, ringed) does not chase the fake down the line; he shuffles along behind the line and stays outside the quarterback. If he had crashed after the fake (dashed), the quarterback would be outside him with nobody in front.

To **set the edge** is to be that last man and win: take on the blocker coming at you (a tight end, a tackle, a pulling guard, a fullback) at or near the line of scrimmage, with your outside arm free, and don't let him push you back or get his head outside you. Then the run has only one place to go, back inside, toward the linebackers and safeties chasing it. A defense talks about a *hard* edge (the defender holds his ground at the line and turns the ball in quickly) and a *soft* edge (he gets pushed back or upfield, and the runner

gets around him). Setting a hard edge is one of the least glamorous and most valued skills an edge player has, and it is a big part of why teams still want 270-pound defensive ends when a 245-pound linebacker might rush the passer just as well. On a given play, the player who turns the ball back inside is also called the *force* defender; who that is, and how it changes with the coverage, is in Run Fits and Box Math.

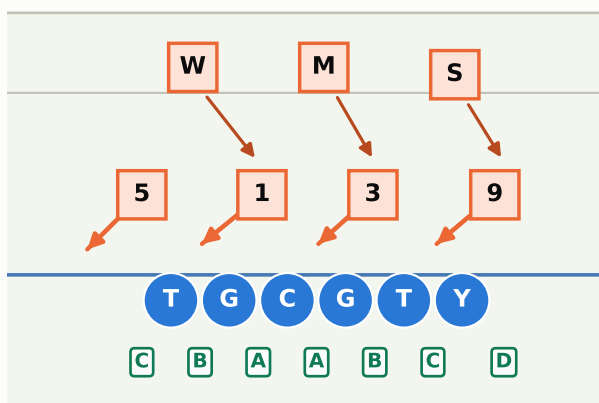
Contain is the edge player's other rule, and its spirit is the same: *nothing gets outside me*, but with an emphasis on patience. On the backside of a run, the end who has contain does not chase the play flat down the line. He trails it at about the depth of the ball, because the plays most likely to come back his way are the ones designed to punish a chasing end: the bootleg, the reverse, the quarterback keeping the ball on a read (Perimeter and Misdirection has those plays from the offense's side). On a pass rush, contain means the ends rush outside the tackles but not past the quarterback's depth. An end who loops too far upfield opens a lane between himself and the quarterback, and a mobile quarterback steps up and runs through it. You will hear analysts call this *rush-lane discipline*; how pass rushers balance it against getting home is in The Pass Rush.

Both jobs ask a fast, aggressive player to be patient on a down when patience isn't rewarded with a statistic. That is exactly why offenses attack them. A defensive end who gets a little too eager for the sack or the tackle for loss is the end the bootleg is built for.

Line movement: slants and angles

So far the linemen have played from where they lined up. But the alignment the offense sees before the snap doesn't have to be the one it blocks after the snap. Defenses can *move* their linemen at the snap, and the two basic movements are the slant and the angle.

Slant (weak): the whole line steps over



Angle: one man, into the next gap

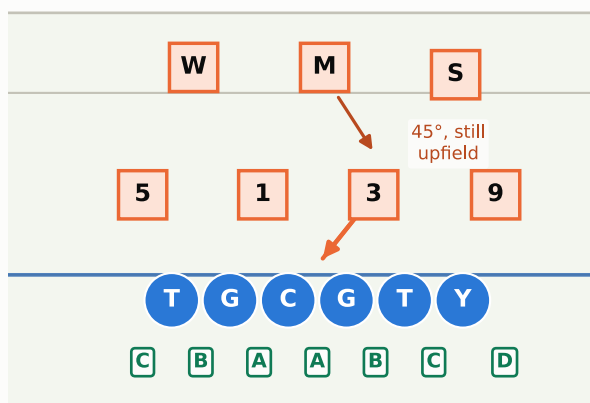


Figure 8: Two ways to move the line at the snap, from the same 4-3 front (pre-snap alignments shown, linemen drawn a little deeper than real so the arrows show; orange arrows are the movement). Left, a slant to the weak side (away from the tight end): every lineman steps hard and flat across the face of the man in front of him into the next gap over, the 9 into the C gap, the 3 into the A gap, the 1 into the left B gap, and the left end widens to keep contain. The linebackers replace them (dashed): the Sam takes the D gap the 9 left, the Mike the right B gap, the Will the left A gap. Right, an angle: just the 3-technique moves, at about 45 degrees, into the A gap, still getting upfield; the Mike replaces him in the B gap. A slant moves the whole picture sideways; an angle moves one or two men while still attacking.

A **slant** is a called movement in which the linemen step hard and flat at the snap, across the face of the blocker in front of them, into the next gap over, usually all in the same direction. The linebackers move the opposite way to fill the gaps the line just left, so the whole front's gap assignments shift by one. An **angle** is a shorter movement, typically by one or two linemen: a step at about 45 degrees into the adjacent gap that moves the defender over while he keeps going *upfield*. Think of the slant as relocating the line, and the angle as a penetrator changing lanes. (The words are not standardized. Some staffs call every movement a slant, some have names for a slant toward the tight end and a slant away from him, and inward and outward moves by pairs of linemen are often called a *pinch* and an *open*. Coaching clinics agree on the mechanics: the first step is a roughly 45-degree step with the back foot, gaining ground and staying low, because a lineman who moves purely sideways can be caught by the blocker and pushed along the line.)¹¹

¹¹On the mechanics (a 45-degree first step, staying low and getting upfield so the blocker can't push a purely lateral slant along the line): Viqtory Sports, "How to coach your players to slant," <https://www.viqtorysports.com/how-to-coach-your-players-to-slant/>. Names for movements vary by staff.

Why move? Because the offense's blocking is decided before the snap. Every offensive lineman has looked at the front and decided whether he is covered or uncovered, who doubles whom and who climbs to which linebacker. Movement changes the answers after it is too late to change the call. A slant can turn a covered lineman uncovered and an uncovered one covered in a single step. A double team aimed at a 3-technique hits air when the 3 angles into the A gap. A pulling guard runs into a defender who wasn't in his path a moment ago. Movement is also how a defense of quick, lighter linemen deals with bigger blockers: they can't win a head-on shove, so they don't take one. As one college staff put it in a coaching clinic, "we use a lot of slants up front (gap control) to try and keep the offensive linemen occupied, which then will allow our linebackers the freedom to roam and get to the ball."¹²

What it costs. Movement is a bet on where the play is going. Slant toward the run and the defense arrives at the hole before the blockers do. Slant *away* from it and every defender has stepped out of the gap the ball is about to come through, and the linebackers who were supposed to replace them have farther to go. Movement is also exactly what zone blocking was built to absorb: zone linemen block whoever arrives in their area, so a slanting defender just runs into the next blocker's arms (Zone Running explains why). Gap schemes, built on pre-snap assignments and pullers, are more exposed to it. And the linebackers have to move with the line every time. If one of them forgets the slant was called, two players fill one gap and the other is empty. When two linemen go further and *loop* around each other on a pass rush, the movement becomes a stunt, which belongs to The Pass Rush.

Putting a front together

Step back and the parts fit together. A defensive coordinator calling a front is choosing, for each lineman, three things you can now name:

1. **Where he lines up** (his technique), which tells the offense who is covered and shapes its blocking.
2. **How he plays it:** one-gap (attack the gap he is in) or two-gap (control the man and read).
3. **Whether he moves:** straight ahead, or a slant or an angle into another gap at the snap.

Different answers to those questions are what separate the fronts in the rest of this part: the even fronts with their 3-techniques and wide ends (Even Fronts), the odd and tite fronts with their 0s and 4is (Odd and Hybrid Fronts), and the linebackers whose jobs change with every one of these choices (Linebackers and the Second Level). A defense can also mix them: two-gap the nose, one-gap the 3-technique, angle the end. The modern NFL front is less a system than a menu.

Film room

i Film room: Tampa Bay Buccaneers, Super Bowl XXXVII (2002 season)

The game: Super Bowl XXXVII, January 26, 2003: Tampa Bay 48, Oakland 21. The Buccaneers' defense, built by Tony Dungy and coordinator Monte Kiffin around a one-gap four-man line and still run by Kiffin in Jon Gruden's first season, intercepted the league MVP, Rich Gannon, a Super Bowl-record five times and returned three of them for touchdowns.¹³ Warren Sapp played the 3-technique, and the design of that whole defense, a fast, light front backed by linebackers who ran to the ball, starts with him.

What to notice:

- Find Sapp before each snap. He moves to whichever side the defense wants its 3-technique; where he lines up tells you which guard has a one-on-one problem.
- His first step. Count how often he is across the line of scrimmage before the guard in front of him has finished his first step. That is one-gap penetration at its purest.
- What happens behind him. When an offense spends a second blocker on him, look for the linebacker that blocker would have taken, and watch him run free.

¹²Matt Welch and Bill Shaw (Friends University), "Defensive philosophy: defensive line movement," Gridiron Strategies: <https://gridironstrategies.com/article/defensive-philosophy-defensive-line-movement>.

¹³"Super Bowl XXXVII," Wikipedia (Tampa Bay 48, Oakland 21, January 26, 2003; Gannon "threw a Super Bowl record five interceptions, three of which were returned for touchdowns"): https://en.wikipedia.org/wiki/Super_Bowl_XXXVII. Kiffin was Tampa Bay's defensive coordinator 1996–2008, under Dungy (1996–2001) and Gruden (2002–08): Buccaneers Ring of Honor, <https://www.buccaneers.com/team/ring-of-honor/monte-kiffin>.

i Film room: New England Patriots, 2004–2010 seasons

The player: Vince Wilfork, a first-round pick in 2004 who played nose tackle in Bill Belichick’s two-gap 3-4 for his first seven seasons: as a rookie he shared the job with the veteran Keith Traylor, and from 2005 it was his alone. (In 2010 he spent some weeks at end, and in 2011 the Patriots went to a four-man front.)¹⁴ He was a Pro Bowler five times and won two Super Bowls, XXXIX and XLIX.

What to do: pick any Patriots early-down base snap from those seasons in which Wilfork lines up over the center. Ignore the ball and watch only him and the center.

- The strike: both hands inside, arms locked, the center held off his body.
- The double team: when a guard comes over, does Wilfork give ground? Usually he doesn’t, and the linebacker behind him is unblocked.
- The shed: watch him throw the blocker to the side the ball isn’t going. Then check the box score; his name often isn’t in it.

i Film room: Los Angeles Rams, Super Bowl LVI (2021 season)

The game: Super Bowl LVI, February 13, 2022, at SoFi Stadium: Rams 23, Bengals 20. Cincinnati got the ball back with 1:25 left needing a field goal to tie. Facing third-and-1 near midfield, the Bengals ran Samaje Perine, and Aaron Donald and nose tackle Greg Gaines tackled him for no gain. On fourth-and-1, Donald got to Joe Burrow and wrapped him up as he threw; the pass fell incomplete, and the Rams won.¹⁵

What to notice: two snaps, one player, both phases of the game. On third down, a run: watch Donald’s get-off and where he is relative to the line when the back arrives. On fourth down, a pass: the same first step, and the same result, a quarterback with no time. It is the best two-play summary of why teams pay a one-gap 3-technique more than almost anyone else on the defense.

i Film room: Philadelphia Eagles 2011

The season: Jim Washburn’s first year coaching the Eagles’ defensive line, and the clearest look at a full-time wide-9 front. The Eagles tied for the league lead with 50 sacks and Jason Babin had 18.¹⁶

What to notice: the ends’ alignment, well outside the tight end on that side (or where one would be), in sprinter’s stances; how quickly they are upfield; and then, on running plays, the space inside them. Watch which linebacker has to fill the lane between the end and the 3-technique, and how often he gets there in time. The pass rush and the run lanes come from the same alignment: that is the trade this chapter has been describing, on the field for a whole season.

One more name belongs here. **Reggie White**, the “Minister of Defense,” played defensive end for the Eagles and the Packers and retired with 198 sacks, then the NFL record (Bruce Smith later passed him). He was a rare end who was both an edge-setter on runs and an overwhelming pass rusher, and in Super Bowl XXXI, after the 1996 season, he sacked Drew Bledsoe three times, a Super Bowl record that has since been tied but never broken, as Green Bay beat New England 35–21.¹⁷ Watch him for what an edge player can be when he never has to choose between the two jobs.

¹⁴Vince Wilfork: 21st overall pick in 2004; Patriots 2004–2014, Texans 2015–2016; Pro Bowls after the 2007 and 2009–2012 seasons; first-team All-Pro 2012; Super Bowls XXXIX and XLIX; shared nose tackle with Keith Traylor as a rookie (and started Super Bowl XXXIX), then played nose for his first seven seasons before New England went to a 4-3 in 2011: https://en.wikipedia.org/wiki/Vince_Wilfork; Pro Football Reference, <https://www.pro-football-reference.com/players/W/WilfVi99.htm>. Traylor’s release in May 2005: Patriots.com, <https://www.patriots.com/news/new-england-releases-dt-traylor-90391>. The two-gap 3-4 and Wilfork as its “two-gapping war daddy”: Chris B. Brown, “Ode to the War Daddies,” Grantland (February 6, 2012), <https://grantland.com/features/bill-belichick-vince-wilfork-new-england-patriots-defense/>.

¹⁵“Super Bowl LVI,” Wikipedia (Rams 23, Bengals 20, February 13, 2022, SoFi Stadium; on third down “Aaron Donald and nose tackle Greg Gaines” tackled Samaje Perine for no gain; on fourth-and-1 Donald “wrapped Burrow up before he could make the throw”): https://en.wikipedia.org/wiki/Super_Bowl_LVI.

¹⁶Same source as the earlier note on the 2011 Eagles; Babin had played for Washburn in Tennessee in 2010. Philadelphia Inquirer, “How Eagles’ Babin became a pass-rushing force” (December 17, 2011): https://www.inquirer.com/philly/sports/eagles/20111217_How_Eagles_Babin_became_a_pass-rushing_force.html.

¹⁷Reggie White: Eagles 1985–1992, Packers 1993–1998, Panthers 2000; 198 sacks (the NFL record when he retired; Bruce Smith later reached 200); three sacks in Super Bowl XXXI, a Super Bowl record since tied by Darnell Dockett (XLIII), Kony Ealy (50) and Grady Jarrett (LI) (Guinness World Records, “Most sacks in a Super Bowl game,” <https://www.guinnessworldrecords.com/world-records/461256-most-sacks-in-a-super-bowl-game>); Hall of

Watch for it on Sunday

Watch for it

Before the snap: find the 3-technique.

- Look at the guards. Is there a defensive tackle on a guard's *outside* shoulder? That is the 3-technique, and his side of the line is where the defense has put its quickness. Is there one big man right over the center? That is a 0 nose, likely playing both A gaps.
- Look at the ends. Shaded on the tackle (a 5)? Way out wide, past where a tight end would be (a wide-9)? Standing up outside the tight end (a 9)? Wider ends mean more pass rush and bigger run lanes inside them.

At the snap: the first step.

- Does the lineman's first step go *into a gap*, upfield (one-gap), or *into the blocker*, with his hands, standing him up (two-gap)?
- Does the whole line step sideways together at the snap? That is a slant; look for the linebackers to shift the other way. One man darting into the next gap while the rest go straight is an angle.

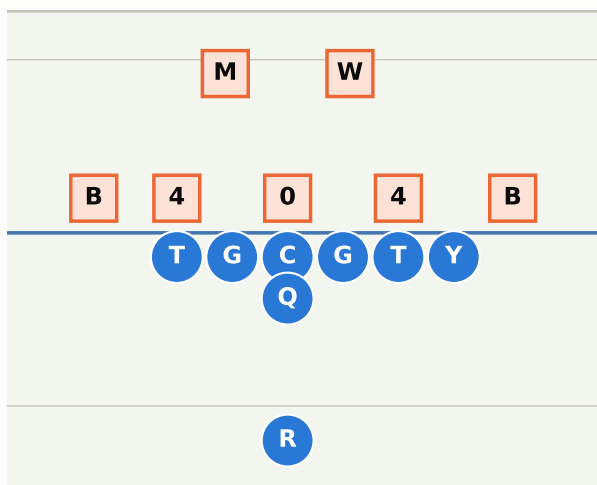
After the play: where did the run go?

- Which way did the run go relative to the 3-technique: at him, or away? Did he get into the backfield, get double-teamed, or get trapped?
- Did the run get outside the edge? If it did, ask who had the edge on that side and whether he got hooked. On a bootleg or scramble, ask who had contain.

For your prediction log (How to Watch a Broadcast): before each early-down snap, note which side the 3-technique is on. After the play, note whether the run went toward him or away from him. A few quarters of that will tell you something about both coordinators.

💡 Predict the play

Front A



Answer 1 is at the end of the chapter. Try it first.

💡 Predict the play

Drill 2. The slant meets the run. First-and-10. The offense runs inside zone to the right. At the snap, the defense's four linemen all slant to the right, toward the tight end, and the picture below is the first half-second. Where is the run most likely to go now, and which defender does the defense need to be right?

Drill 2: inside zone right meets a slant right

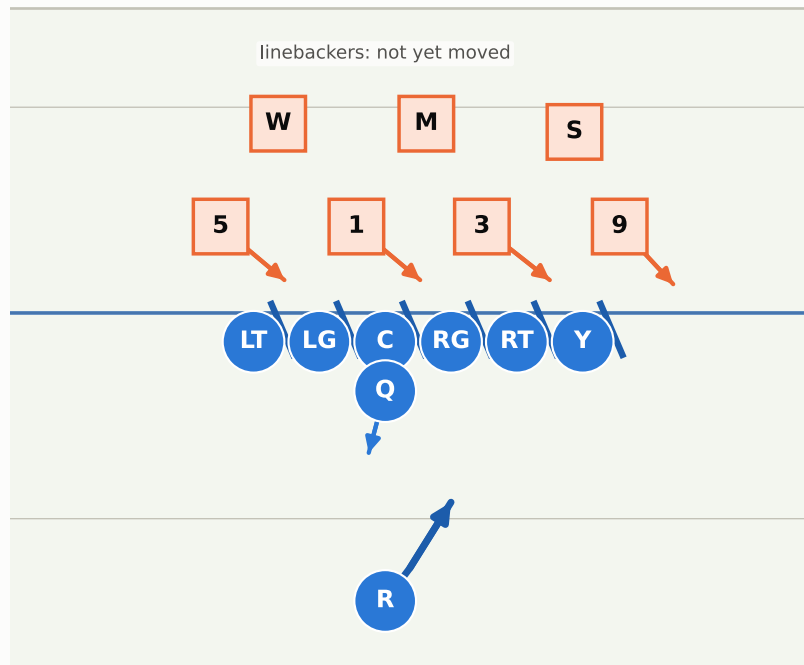


Figure 10: Drill 2: the first half-second. The offense's line steps right for inside zone and the back aims at the right guard's outside hip; the defense's linemen (orange arrows; drawn a little deeper than real) all slant hard to the right at the snap, the end outside the tight end widening. The linebackers have not moved yet.

Answer 2 is at the end of the chapter. Try it first.

💡 Predict the play

Drill 3. Who has contain? Third-and-7. The offense is in the shotgun with an empty-looking backfield and a quarterback who likes to run. The defense rushes four: a wide-9 on the right, a 5 on the left, and two tackles inside. Which rushers have contain, how should they rush, and where is the quarterback most likely to escape if somebody gets it wrong?

Drill 3: four rushers, one mobile quarterback

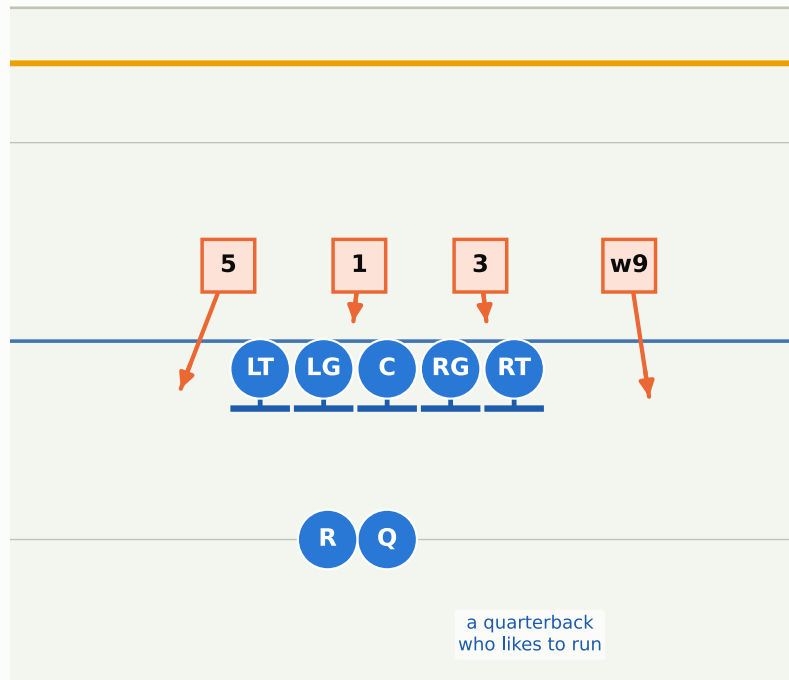


Figure 11: Drill 3: third-and-7, shotgun, a mobile quarterback. The defense rushes four (orange arrows show the first steps; linemen drawn a little deeper than real): the wide-9 on the right, the 5 on the left, a 3 and a 1 inside. Everyone else drops into coverage (not shown).

Answer 3 is at the end of the chapter. Try it first.

Takeaways

- **Alignment is a job description.** The nose (0 or 1) anchors against double teams and wants mass; the 3-technique gets a one-on-one with a guard and wants quickness; the 4i and 5 want length to control a tackle; the 9 and the wide-9 want speed, and the wider the end, the more pass rush and the bigger the run lane inside him.
- **One-gap attacks.** Each lineman owns one gap and attacks it, aiming for penetration. It wins with quickness, creates sacks and tackles for loss, and turns the linemen loose, but a wrong guess leaves an empty gap, and traps, draws and screens are built to punish it.
- **Two-gap controls.** A head-up lineman strikes the blocker, locks out, reads (the blocker's head tells him where the ball is going) and sheds to the ball's side. One man covering two gaps frees the linebackers, at the cost of rare bodies, little pass rush, and a stat sheet that hides his value.
- **Fight pressure with pressure.** Against a down block, the man blocked fights back outward and the man left alone squeezes and looks for the puller; against a reach, step with the blocker and keep the outside arm free; against a double team, give no ground and keep both blockers.
- **The edge is everybody's boundary.** Setting the edge turns runs back inside; contain keeps the quarterback and the misdirection plays from getting outside. Both ask an aggressive player for patience.
- **Movement changes the picture after the snap.** Slants shift the whole line a gap; angles move one man while he keeps attacking. Both beat pre-snap blocking assignments and both are bets on direction that the linebackers have to back up.

Connections

Builds on: The Twenty-Two (the nose tackle, defensive tackles and ends, edge rushers), A First Look at Defense (the 4-3, the 3-4 and the box) and The Language of the Run Game (gaps, technique numbers and the blocks), with the blocking schemes from Zone Running and Gap and Power Runs as the opponents.

Sets up:

- Even Fronts: the 4-3 Over and Under, the wide-9 front and the 46, built from the 3-technique, the 1 and wide ends.
- Odd and Hybrid Fronts: the 3-4, the tite front's 4i-0-4i, and why two-gap ideas came back in the light-box era.
- Linebackers and the Second Level: the players who fill the gaps the line leaves, and replace them after a slant.
- Run Fits and Box Math: spill and squeeze, force, and how eleven gap assignments add up.
- The Pass Rush: the same linemen on passing downs, rush moves, stunts, and why pressure matters more than sacks.

See also (case study): Fangio's Two-High Revolution and Macdonald's Answer, for what happens to defensive line play when a defense keeps two safeties deep and asks six men to hold the box.

If you want to go deeper: Pro Football Focus's "Defensive Prototypes" series (one article per technique, with player examples); Chris B. Brown's *The Art of Smart Football* on how fronts and run schemes answer each other; and any Patriots game from the late 2000s and any Rams game from 2018, watched for nothing but the man over the center and the man outside the guard.

i Go deeper

Penetration, measured. Tracking data turns this chapter's distinction into a number. A one-gap penetrator is *behind* the line of scrimmage a second after the snap; a two-gapper is *at* it. In the two simulated plays above, here is how far each ringed defender is past the line (negative numbers mean he is in the offense's backfield), computed from their Big Data Bowl-format tracking frames exactly as you would from real data:

	defender	+0.3 s	+0.6 s	+1.0 s
0	one-gap 3-technique (simulated)	0.0	-1.4	-2.6
1	two-gap nose (simulated)	0.7	0.6	0.2

The same calculation on real frames is a few lines with `gridiron.bdb`. The 2025 Big Data Bowl release (2022 season, weeks 1-9) has every player's location ten times a second, and its players file has his position label, so you can find each interior defensive lineman on each run and measure how deep he is at one second. Aggregated by player, it separates the penetrators from the anchors. The data needs a free Kaggle login, so this cell is not run when the book is built.

Answers to Predict the play

Answer 1.

Front A looks built to two-gap; Front B looks built to one-gap. The clue is the alignments. Every lineman in Front A is *head-up* on a blocker (0, 4, 4), and a head-up alignment doesn't put a defender in either gap, so the natural job is to control the man and take whichever side the ball goes: the nose both A gaps, each end the B and C gaps on his side. Expect big bodies: a nose well over 320 pounds, long ends near 300. The free players are the two inside linebackers, who have no gap to rush to at the snap and can run to the ball, as long as the three big men keep all five offensive linemen busy.

Every lineman in Front B is *shaded*, sitting in a gap (5 in the C, 1 in the A, 3 in the B, 9 in the D), which is what a one-gap front looks like: each attacks the gap he is in and the three linebackers take the three that are left (left B, right A, right C). Expect a quick 3-technique, lighter ends and more penetration. Nobody in Front B is "free"; everybody has a gap, and the defense's bet is that each of them wins his.

The catch: alignment is a strong hint, not a guarantee. Wade Phillips ran a one-gap defense out of 3-4 looks for decades, and many 4-3 teams ask a 1-technique to play both A gaps. Confirm it with the first step after the snap: into the gap, or into the man.

Answer 2.

Most likely back to the left, on the cutback. The slant went the same way as the play, so the play side is crowded: each defensive lineman has stepped across the face of a blocker into the next gap toward the tight end, and the gaps the back was aiming at (the

right A and B) now have defenders arriving in them. Zone blocking absorbs some of that (each lineman simply blocks whoever came into his area), but the back reads a crowded play side and does what inside zone teaches him: he *bends* it back toward the gaps the slant just emptied, the left A and B.

The defender who has to be right is the Will (the backside linebacker). On a slant the linebackers replace the gaps the line left, and the backside gaps are the Will's. If he remembered the call and filled, the back cuts back into him for a short gain, and the slant won. If he flowed right with everything else, the cutback lane is empty and the next defender is a safety. That is the bet every slant makes: it is great when it guesses the direction, and the linebackers have to clean up when it doesn't.

Answer 3.

The two ends have contain: the 5 on the left and the wide-9 on the right. Their rule is to rush *outside* the tackles but not *past* the quarterback's depth. If either one runs a wide arc around the tackle and keeps going, he ends up behind the quarterback, and the space between him and the defensive tackle next to him becomes a running lane. A mobile quarterback steps up into the pocket and out through that lane, usually for a first down on third-and-7.

The wide-9 is the likeliest to get it wrong, because his whole alignment is built for a long, fast arc to the quarterback. Against a runner he has to flatten his rush earlier and stay level with the quarterback. The two inside rushers have the opposite job: push the middle of the pocket back into the quarterback so he can't step up. A four-man rush against a mobile quarterback is a box: two sides that must hold, and a front that must squeeze. When one side bursts, look for the quarterback there. The Pass Rush has the rest of this, including the stunts that trade one rusher's lane for another's.